Computer-Aided Diagnosis of Lung Cancer via Transthoracic Ultrasonography

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***Abstract*—The integration of computer-aided diagnosis (CAD) systems in clinical practice is becoming essential for enhancing the precision and objectivity of medical image interpretation. This paper proposes an automated system for lung cancer diagnosis using transthoracic ultrasound (TUS) imaging. By integrating advanced image processing for lesion segmentation and logistic regression for classification, the system extracts quantitative features to support clinical decision-making and reduce diagnostic subjectivity. Experimental results demonstrate that the inclusion of an uncertainty-aware logic provides a reliable second opinion, effectively flagging borderline cases for expert review to ensure clinical safety and diagnostic reliability.**

***Keywords*—Transthoracic ultrasound, lung cancer, image segmentation, computer-aided diagnosis, logistic regression, speckle noise reduction.**

I. Introduction

Lung cancer remains a primary cause of global mortality, requiring early and accurate diagnosis to improve survival rates. As documented in [1], lung cancer continues to lead global cancer statistics in terms of incidence and mortality. While medical imaging has evolved, the need for objective diagnostic tools remains critical. Computer-aided systems, as discussed in [2], can reduce the subjectivity inherent in manual image evaluation, providing quantitative indicators for clinical support.

The novelty of this work lies in the development of a fully autonomous diagnostic framework for ultrasound imaging. Unlike existing semi-automated approaches, this system utilizes a multi-level segmentation logic and an uncertainty-aware classification model. By introducing an "Uncertain" category, the system avoids forced binary decisions, reflecting the caution required in real-world clinical practice.

A. Diagnostic Gold Standards

In current medical practice, Computed Tomography (CT) and Magnetic Resonance Imaging (MRI) represent the gold standard for diagnosing and evaluating pulmonary lesions. These modalities provide high-resolution images that allow for precise anatomical localization. However, they are characterized by significant drawbacks, including high operational costs, lack of portability for bedside monitoring, and, in the case of CT, exposure to ionizing radiation.

B. Current State of the Art in CAD Systems

Transthoracic Ultrasonography (TUS) has emerged as a reliable "State of the Art" alternative for real-time, non-invasive imaging. While TUS is cost-effective and portable, its primary challenge lies in the presence of speckle noise and low contrast between tissues. Existing computer-aided diagnosis (CAD) systems often attempt to overcome these hurdles through semi-automated methods. These systems frequently require clinical experts to manually initialize markers for the region of interest or the tumor boundaries. While such tools improve objectivity, they still suffer from inter-observer variability and remain time-consuming for the practitioner.

C. Proposed Autonomous Solution

In contrast to traditional semi-automated approaches, including the methodology discussed in [3], the solution proposed in this work transitions toward a fully autonomous pipeline. By replacing manual initialization with a multi-level automated segmentation logic, the system ensures high consistency across different datasets. Furthermore, unlike forced-choice binary models, this approach integrates an uncertainty-aware classification that better reflects the caution required in clinical diagnostic environments.

The remainder of this paper is structured as follows: Section II details the system design and implementation, covering preprocessing, autonomous segmentation, and the classification model. Section III presents the results analysis, including quantitative metrics and inference case studies. Finally, Section IV provides the conclusions and future directions.

II. System Design and Implementation

A. Dataset Characteristics

The dataset utilized in this research consists of 203 2D grayscale ultrasound images. As emphasized in [4], the use of accurately pre-labeled datasets is essential for the validation of automated medical analysis systems, allowing for a direct comparison between the system's output and the actual clinical diagnosis (malignant vs. non-cancerous).

However, pulmonary ultrasound images present significant technical challenges. According to [5], the high variability and low contrast of these images significantly hinder automated processing. These difficulties are further detailed in [6], which notes that malignant masses often exhibit irregular margins and spiculations. Furthermore, the presence of speckle noise, inherent to ultrasound physics, obscures structural clarity and complicates the precise delimitation of tumor boundaries. Consequently, these factors justify the necessity for a robust

pre-processing stage to isolate the Region of Interest (ROI) from adjacent anatomical structures.

B. Preprocessing and ROI Extraction

The preprocessing stage is essential for mitigating the inherent artifacts of ultrasound imaging and isolating the relevant anatomical structures for analysis. Our pipeline is structured into two primary phases:

First, to address the characteristic speckle noise while preserving critical edge information, several spatial filters were evaluated. The *Gaussian filter* was selected as the optimal solution, as it provided the most effective balance between noise suppression and the retention of lesion contours. Following filtration, global histogram equalization was applied to redistribute intensity levels. This step is crucial for enhancing the contrast between the pulmonary lesion and the surrounding lung parenchyma, facilitating more accurate segmentation, as illustrated in Fig.1.

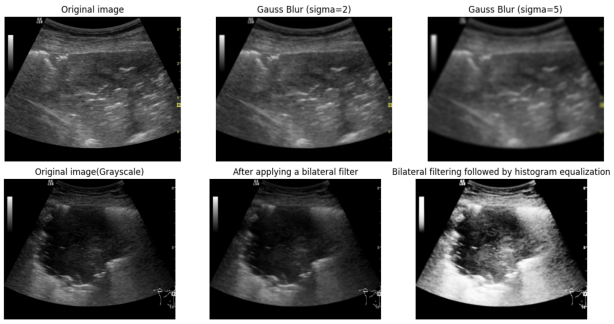


Fig. 1. Image enhancement results: (top) Gaussian filtering for speckle noise reduction; (bottom) Final contrast enhancement via histogram equalization.

Second, a fully automated masking technique was developed to isolate the Region of Interest (ROI). While initial experimental attempts utilized simple linear delimiters for the lateral margins, they lacked the robustness required for varied datasets. The final implementation employs a multi-step morphological approach to ensure consistent results:

- Automated binarization using Otsu's thresholding to separate the ultrasound field from black margins and metadata;
- Morphological "closing" operations to bridge discontinuities and stabilize the boundary;
- Selection of the external contour with the largest area to ensure the final ROI corresponds strictly to the ultrasound cone.

The effectiveness of this autonomous extraction across different cases from the dataset is illustrated in Fig. 2.

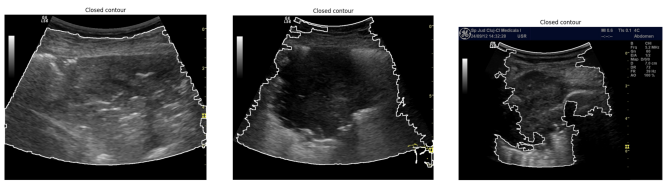


Fig. 2. Examples of automated ROI extraction: final stabilized ultrasound field contours obtained for different images in the dataset.

C. Segmentation Logic: From Semi-Automated to Autonomous

The core of the diagnostic system lies in the precise delimitation of pulmonary lesions within the previously extracted ROI. This process transitioned from a user-guided approach to a fully autonomous framework to ensure diagnostic objectivity.

1) *Semi-automated User-Guided Segmentation*: Initially, a semi-automated pipeline was implemented, combining *Watershed* segmentation with *Active Contour (Snake)* models. All processing is restricted to the ROI mask to avoid background interference. The user interaction is based on a keyboard-driven interface where keys *1* through *9* allow the placement of multi-colored markers. In clinical practice, the following primary seeds were utilized:

- *Key 1 (Green Marker)*: Used to identify the tumor region (hypoechoic zone).
- *Key 2 (Red Marker)*: Used to identify healthy tissue or the ultrasound background.

Upon triggering the algorithm, the Watershed transformation generates a region map based on these seeds. The segmentation is subsequently refined using the Snake method. However, as illustrated in Fig. 3, this stage proved sensitive to seed density and placement.

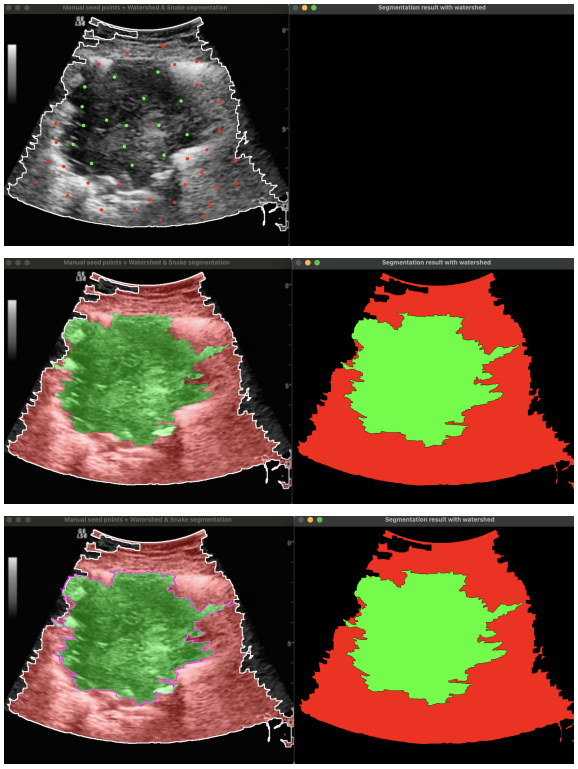


Fig. 3. Semi-automated segmentation flow: (top) Manual markers; (middle) Watershed regions; (bottom) Final Snake refined contour.

2) *Autonomous Multi-Level Segmentation*: To eliminate subjectivity, a fully autonomous algorithm was developed. The process begins with *Multi-level Otsu thresholding*. This technique splits the image into three distinct intensity zones, automatically selecting the low-intensity class as the tumor candidate.

In parallel, the system identifies *hyperechoic linear structures* (highlighted as blue contours in the output), which represent bright anatomical features or artifacts within the ROI. The resulting autonomous pipeline produces four key outputs: the *ROI mask*, the *tumor mask*, the *detected linear structures*, and a *final overlay image* (as seen in Fig. 4) that combines all information for clinical review.

The tumor mask is refined using morphological operations to isolate the dominant connected component, and the Active Contour model is initialized to stabilize the boundary. This method provides consistent results across the dataset.

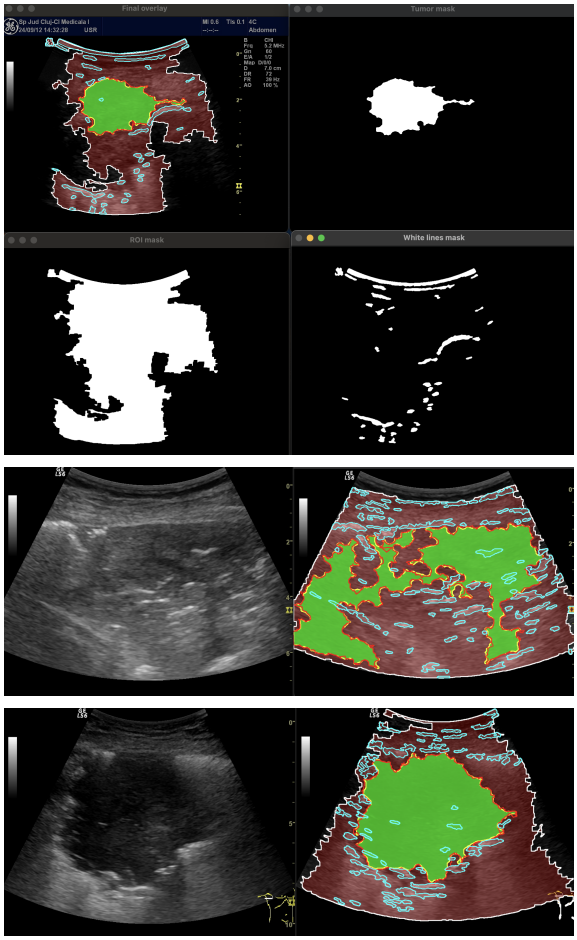


Fig. 4. Autonomous results and multi-mask output.

D. Classification Model and Experimental Environment

The system was fully implemented within the *Google Colab* cloud environment, accessing 203 ultrasound images from a *Google Drive* repository. For the final classification stage, we employed a *Logistic Regression* model, partitioned using a 70/30 stratified split.

The model processes a 9-feature vector extracted from the segmented masks, capturing critical geometric and intensity-based markers. These numerical descriptors, illustrated in the feature matrix in Fig. 5, provide the quantitative foundation for the automated diagnosis. To evaluate the system's performance, we defined the following metrics:

- *Accuracy*: The proportion of total correct predictions.
- *Sensitivity (Recall)*: The ability to correctly identify malignant cases.
- *Precision*: The accuracy of the positive predictions.

```
Total images: 203
Features used: ['roi_ratio', 'tumor_ratio_in_roi', 'elongation', 'circularity', 'mean_roi', 'mean_tumor', 'white_below', 'white_above', 'n_wh
   label  roi_ratio  tumor_ratio_in_roi  elongation  circularity   mean_roi  \
0      1   0.307769            0.111913    1.821918    0.289692  74.091296
1      1   0.357299            0.076961    1.847414    0.884821  64.073233
2      1   0.122740            0.235234    1.605634    0.117726  74.510776
3      1   0.511669            0.152749    1.811715    0.161017  53.335879
4      1   0.342783            0.184201    1.981235    0.178476  72.092127

   mean_tumor  white_below  white_above  n_white_cnts
0   44.431763     0.046191     0.013723            25
1   38.893451     0.030135     0.021634            29
2   62.274718     0.010242     0.062969            20
3   41.598341     0.006852     0.064984            71
4   75.665177     0.024423     0.036953            32
```

Fig. 5. The extracted feature matrix for the analyzed cases.

E. Interface Design

The final component of the system design is the graphical user interface (GUI), developed using the Gradio framework to facilitate real-time interaction between the clinician and the autonomous pipeline. The interface serves as a functional prototype for a Computer-Aided Diagnosis (CAD) workstation, designed to be intuitive for medical personnel without requiring programming knowledge.

As illustrated in Fig. 6, the dashboard is divided into an input area for raw ultrasound image upload and a dual-output results panel. Once an image is submitted, the backend executes the full processing sequence—from ROI extraction to logistic regression classification.

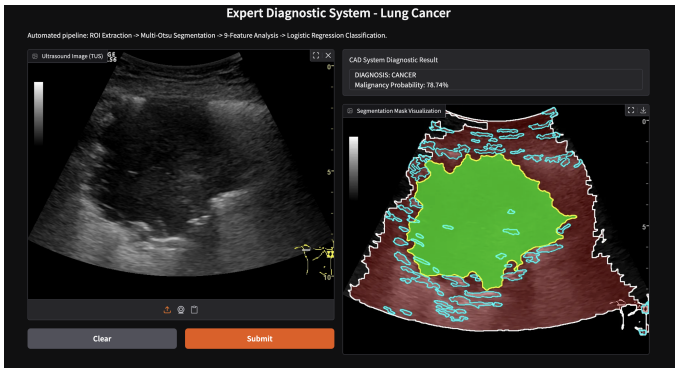


Fig. 6. The developed diagnostic interface showing the raw ultrasound input (left) and the real-time autonomous processing results with probabilistic verdict (right).

III. Results Analysis

This section details the experimental validation of the CAD system, moving from the conceptual design to the actual performance on the clinical dataset.

A. Quantitative Performance and Classification Metrics

The model's performance was evaluated using the test set. We extracted the confusion matrix and the classification report to maintain diagnostic transparency, as illustrated in Fig. 7.

The results indicate a *high sensitivity (Recall = 0.76)* for the cancer class. Given the dataset of 203 images, these metrics provide a statistically significant evaluation of the system's diagnostic capabilities. The "Uncertain" logic effectively flags borderline cases, ensuring that ambiguous intensity profiles are prioritized for clinical review, thereby maintaining a high level of diagnostic reliability across diverse clinical cases.

```
Confusion matrix:
 [[ 8 16]
 [ 9 28]]

Raport:
              precision    recall  f1-score   support

           0       0.47      0.33      0.39        24
           1       0.64      0.76      0.69        37

    accuracy                           0.59        61
   macro avg       0.55      0.55      0.54        61
weighted avg       0.57      0.59      0.57        61
```

Fig. 7. Classification results: Confusion Matrix and metrics.

B. Inference Case Studies

The system's response to various clinical profiles is illustrated in Fig. 8, covering malignant, non-cancerous, and borderline cases. The autonomous pipeline provides accurate visual representations of lesion boundaries, while the classifier calculates malignancy probabilities to assist the final diagnostic decision. To ensure clinical safety, an "Uncertain" verdict is triggered if the relative lesion area within the region of interest is below 0.25 or if the probability score falls within a "Safety Zone" of 0.49–0.51. This logic effectively flags cases with insufficient visual evidence or ambiguous statistical profiles for expert review.

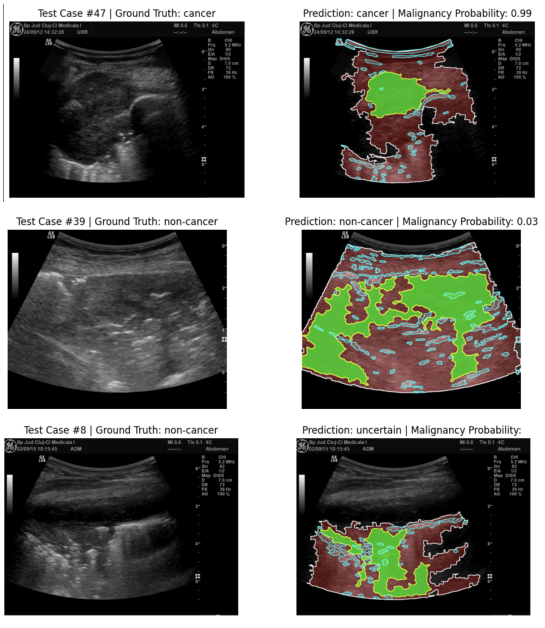


Fig. 8. System inference results: (top) Correct identification of a malignant lesion; (middle) Correct identification of a non-cancerous case; (bottom) Case categorized as "Uncertain" for clinical review.

IV. Conclusions

In this research, a complete autonomous system was developed for the analysis and classification of pulmonary ultrasound images. The proposed framework effectively differentiates between malignant and non-cancerous cases based on automatically extracted features, as seen in the experimental results.

By transitioning from a semi-automated approach to a fully automated pipeline utilizing Multi-level Otsu thresholding and Active Contours, the dependency on user intervention was eliminated. As suggested in [7], automated segmentation is a critical step in reducing observer variability and enhancing the clinical utility of ultrasound imaging.

The results demonstrate the efficacy of the 9 extracted quantitative features in providing a robust basis for malignancy identification. The inclusion of a "Safety Zone" for borderline cases reflects a realistic clinical diagnostic process. This approach aligns with the findings presented in [8], which suggests that computer-aided systems should function as "second opinions" that prioritize reliability by flagging uncertain cases for expert review.

As a primary direction for future development, the implementation of a *closed-loop validation mechanism* is proposed. This would involve an iterative logic where the model's output is cross-verified or weighted against a secondary architecture, further reducing classification errors and increasing the overall reliability of the diagnostic verdict.

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References

[1] H. Sung et al., "Global Cancer Statistics 2020: GLOBOCAN Estimates of Incidence and Mortality Worldwide for 36 Cancers in 185 Countries," *CA: A Cancer Journal for Clinicians*, vol. 71, no. 3, pp. 209-249, 2021.

[2] K. Doi, "Computer-aided diagnosis in medical imaging: Historical review, current status and future potential," *Computerized Medical Imaging and Graphics*, vol. 31, no. 4-5, pp. 198-211, 2007.

[3] C. Achim, R. Rusu-Both, E. H. Dulf, and R. I. Chira, "Lung Cancer Diagnosis based on Ultrasound image processing," in *2018 22nd International Conference on System Theory, Control and Computing (ICSTCC)*, Sinaia, Romania, 2018.

[4] W. M. Hafizah and E. Supriyanto, "Development of 3D image reconstruction based on unguided 2D fetal phantom ultrasound images using VTK," *WSEAS Transactions on Signal Processing*, vol. 6, no. 10, pp. 198-207, Oct. 2010.

[5] R. Chira, A. Chira, and P. A. Mircea, "Intrathoracic tumors in contact with the chest wall - ultrasonographic and computed tomography comparative evaluation," *Medical Ultrasonography*, vol. 14, no. 2, pp. 115-119, 2012.

[6] Mitravathi, S. Murali, and M. S. Dinesh, "Classification of Mass in Breast Ultrasound Images using Image Processing Techniques," *International Journal of Computer Applications*, vol. 42, no. 10, Mar. 2012.

[7] J. A. Noble and D. Boukerroui, "Ultrasound image segmentation: a survey," *IEEE Transactions on Medical Imaging*, vol. 25, no. 8, pp. 987–1010, 2006.

[8] J. Shiraishi et al., "Computer-aided diagnosis and decision support systems in medical imaging," *European Journal of Radiology*, vol. 78, no. 2, pp. 210-215, 2011.

[9] scikit-image Developers, "Active contour model (snake)," *scikit-image documentation*, version 0.25.x. [Online]. Available: https://scikit-image.org/docs/0.25.x/auto_examples/edges/plot_active_contours.html. [Accessed: May 2026].

[10] GeeksforGeeks, "Contour Detection with Custom Seeds using Python-OpenCV," 2023. [Online]. Available: https://www.geeksforgeeks.org/python/contour-detection-with-custom-seeds-using-python-opencv/. [Accessed: May 2026].